Genome–wide association analysis of yield reduction under water deficit stress reveals key loci association with drought tolerance in soybean

Simon Lackey[1,2], Nayla Zalzalah[1,2], Nathaniel Ort[1], Jana Biova[3], Maria Skrabisova[3], Elroy Cober[1], Ashkan Golshani[2], Malcolm Morrison[1], Bahram Samanfar[1,2]

SamanfarLab Functional Genomics

Carleton University

[1]Agriculture and Agri-Food Canada, 960 Carling Ave, Ottawa, Canada
[2]Department of Biology, Ottawa Institute of Systems Biology, Carleton University, Ottawa, Canada
[3]Department of Biochemistry, Faculty of Science, Palacký University in Olomouc, Czech Republic.

Agriculture and Agri-Food Canada

Agriculture et Agroalimentaire Canada

Introduction:

Soybean is a globally important crop contributing to food security through high seed protein and oil content and environmental sustainability thanks to its symbiotic nitrogen fixation [1]. Water deficit stress and drought during the growing season is increasing with climate change, and can have a significant impact on yield in almost all growing regions even when total precipitation is adequate but inconsistently delivered throughout critical yield producing physiological periods [2]. Selection of drought tolerant genotypes for crop breeding is an emerging and critical challenge relying on high quality phenotypic assessment as well as modern genomics techniques such as genome-wide association study (GWAS).

Methods:

A panel of Canadian short season soybean genotypes (n=137) was grown in innovative side by side irrigated and non-irrigated treatment yield trial experiments in 2018, 2019, 2021, and 2022 in Ottawa, Ontario, Canada. Phenotypes representing tolerance to water deficit stress and drought were generated by comparing genotype performance between the two treatment conditions, such as yield and yield components (ΔYLD%, ΔSDm2, ΔSDm2%, ΔSDWT, ΔSDWT%) and seed composition (ΔP, ΔP%, ΔO, ΔO%). ANOVA was carried out in R [3] to assess normality and multi-locus GWAS was carried out using 19139 polymorphic SNPs and the mrMLM v4.02 GUI and package [4,5]. Haplotype blocks were delineated using Haploview 4.1 [6], and allele substitution effect was calculated [7]. Post-GWAS analysis was carried out using the Accuracy approach [8], the AccuCalc package [9], and the Genomic Variation Explorer [10].

Acknowledgements:

This study would not have been possible without the technical assistance provided by the ORDC soybean phenomics, genomics, and breeding teams.

CFCRA CANADIAN FIELD CROP RESEARCH ALLIANCE | ARCCC ALLIANCE DE RECHERCHE SUR LES CULTURES COMMERCIALES DU CANADA

UNIVERSITÉ LAVAL | GenomeQuébec

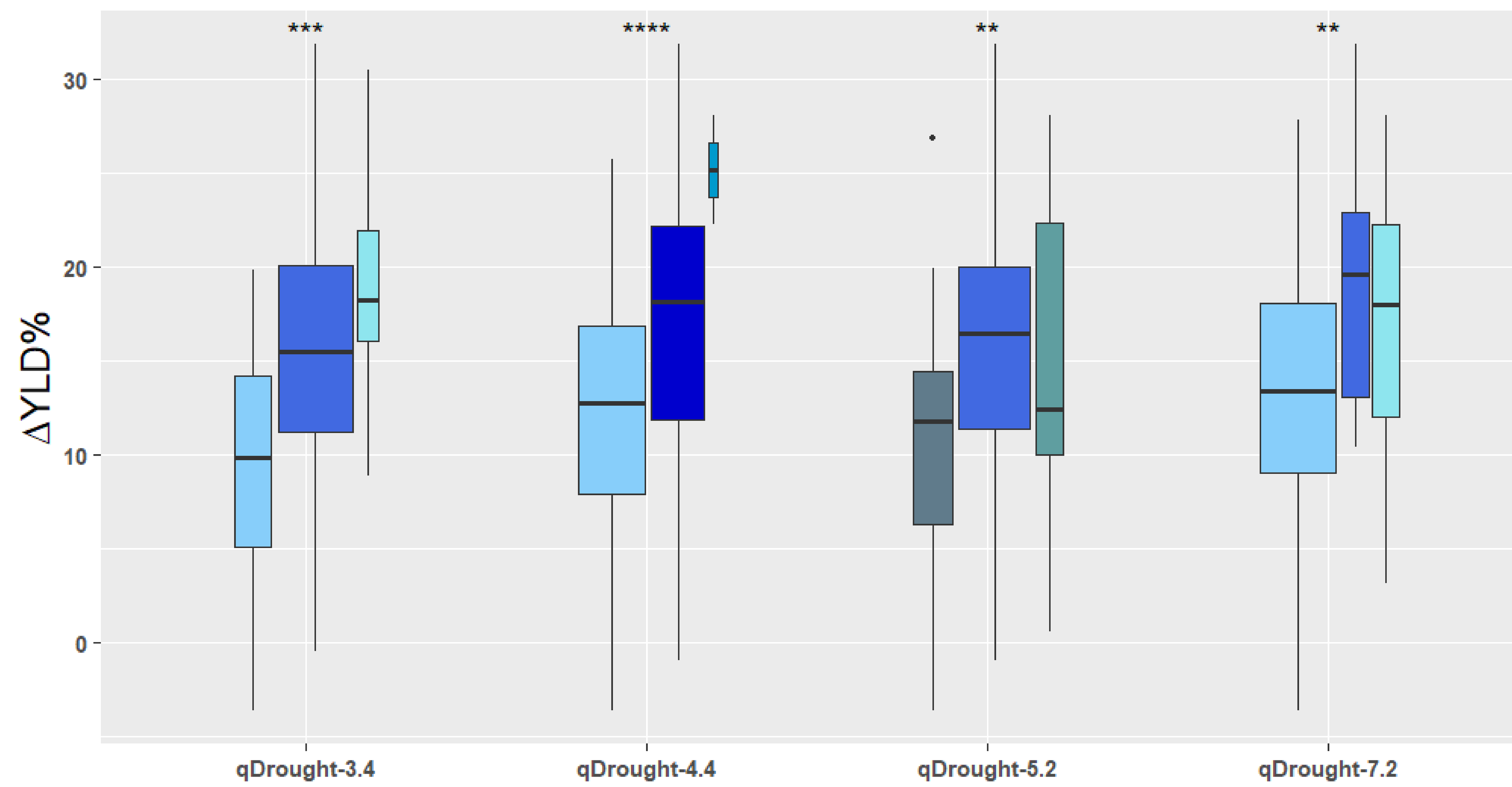

Figure 1: ASE for ΔYLD% detected in short season Canadian soybean tested under irrigated and non-irrigated treatments.

Results:

Irrigation of soybean yield trial plots delivered between 29% and 42% additional water availability compared to the non-irrigated treatment over the four years studied. However, this increase in water availability resulted in a range in yield loss between 0% and 30%, indicating that genotypic response is not uniform. The effect of irrigation on phenotypic response was significant for yield, seed weight, seeds per m^2, and oil, but not significant for seed protein content. Association analysis revealed associations on chromosomes 3, 4, 5, 6, 7, 8, 14, 16, and 20 for traits derived from comparison between treatments. Four QTL for ΔYLD% (*qDrought-3.4, qDrought-4.4, qDrought-5.2, and qDrought-7.2*) have ASE between 4.64% and 6.32% yield loss between allelic classes. Post-GWAS analysis for ΔYLD% QTL revealed important variants, one of which, *Glyma.03G25*****, has high impact variants in the promoter region that affect the binding of important soybean transcription factors.

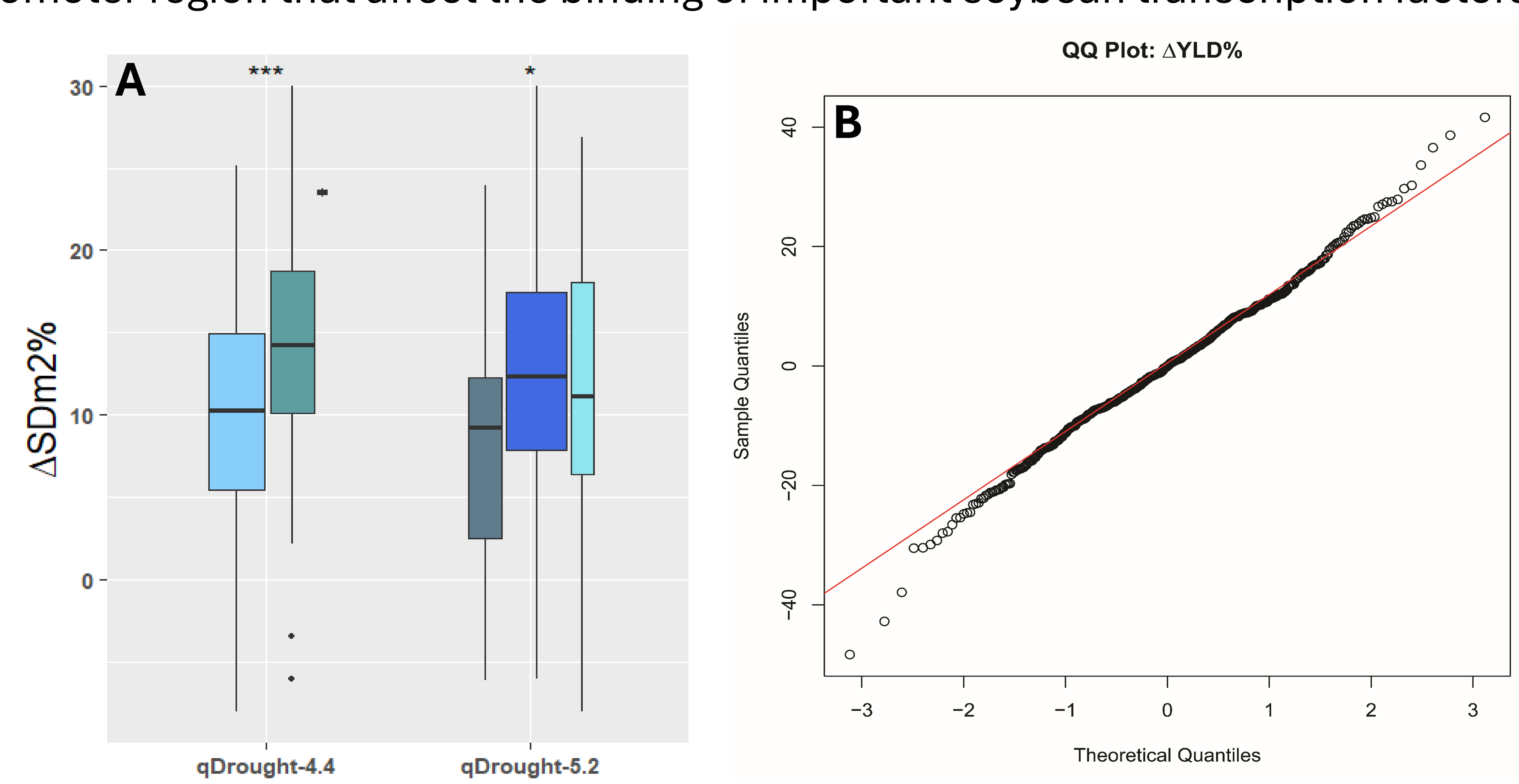

Figure 2: A) ASE for ΔSDm2% detected in short season Canadian soybean tested under irrigated and non-irrigated treatments. **B)** Phenotypic QQ plot of ΔYLD% showing normality of the data.

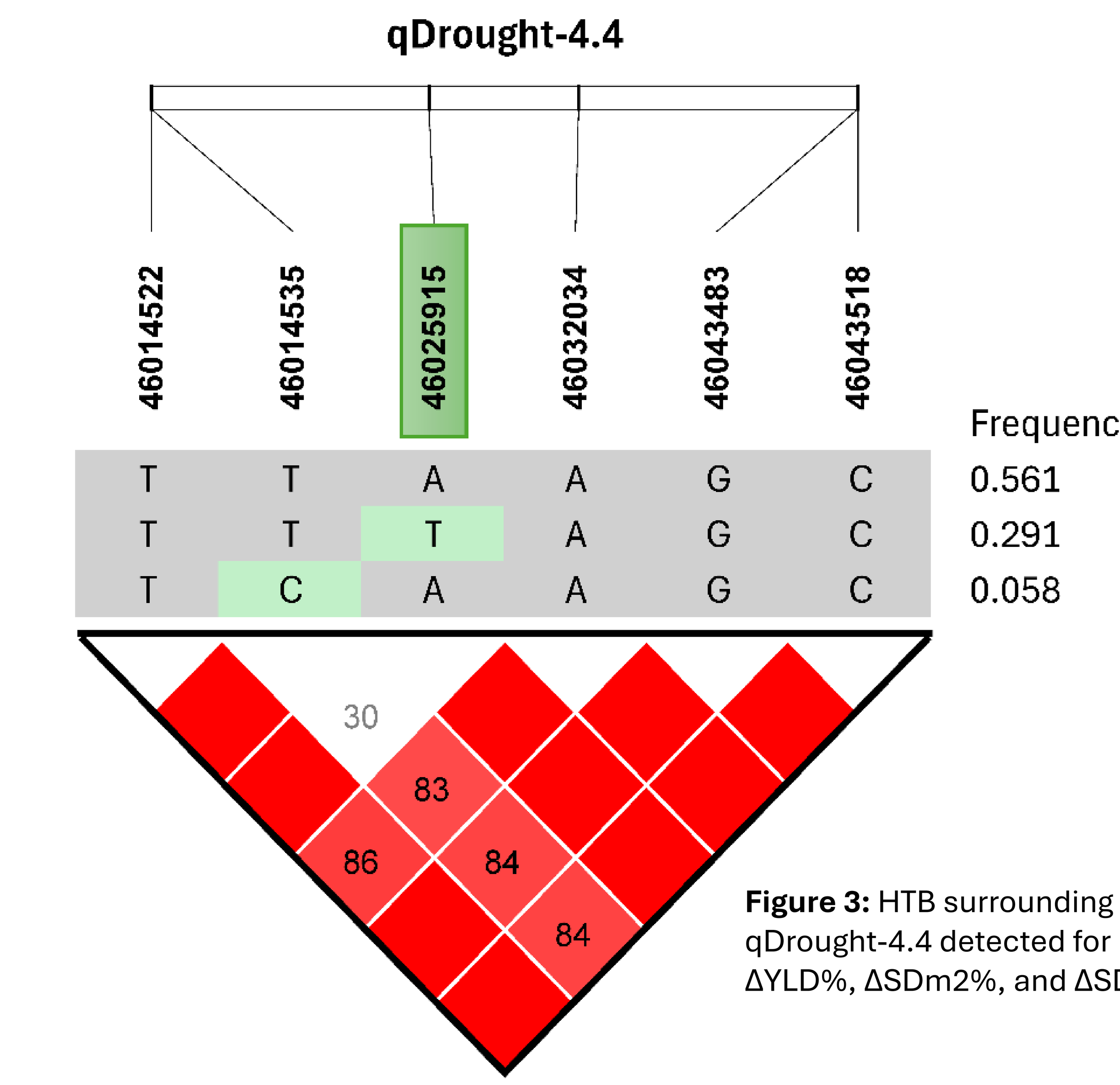

Figure 3: HTB surrounding qDrought-4.4 detected for ΔYLD%, ΔSDm2%, and ΔSDm2 .

Discussion and Conclusion:

Climate change is predicted to increase the incidence of drought and cause between 7% and 23% in crop losses across regions and species and would require costly changes in crop inputs to mitigate [11]. Spring planted soybean in temperate regions are susceptible to water deficit stress as their yield production is largely controlled by moisture availability during critical pod filling periods in hot and dry summer months. Reducing yield penalty due to water deficit stress by breeding for enhanced climate resilience will be a key factor in ensuring the ongoing success of Canadian soybean production. Pyramiding of QTL such as the four detected in this study for tolerance to water deficit stress in desirable genetic backgrounds could prove foundational for the next generation of crop breeding. Understanding of molecular pathways, such as changes in transcription factor binding for UDP-glycosyltransferase genes (*Glyma.03G25*****), and how they can be manipulated to combat climate change, represents a major avenue for ongoing study in crop science and beyond.

Digital Poster and References: